

# Comment on "Dynamic Control of Cardiac Alternans"

In a recent Letter, Hall *et al.* [1] demonstrated that a feedback control method can be used to suppress alternans rhythms in a piece of cardiac tissue. Here we point out that the method employed is properly viewed as a modification of a previously analyzed time-delay feedback method and that comparison of the two immediately yields a surprising and potentially significant result.

From the perspective of linear control theory, the system studied by Hall *et al.* can be modeled as a generic dynamical system with a return map having an unstable fixed point  $X^*$ . In the linear vicinity of  $X^*$  the equation governing the system with feedback applied as

$$X_{n+1} = -\mu X_n + \delta \lambda_n, \quad (1)$$

$$\delta \lambda_n = \Theta_n \beta (X_n - X_{n-1}), \quad (2)$$

$$\Theta_n = \begin{cases} 1 & \text{if } \beta(X_n - X_{n-1}) > 0, \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

where in terms of the parameters of Ref. [1] we have  $\mu = A$  and  $\beta = \alpha B/2$ . (Equations (1)–(3) are equivalent to Eq. (2) of Ref. [1] in the linear vicinity of  $X^*$ , with  $\Theta_n$  restricting the application of control to positive  $\delta \lambda_n$  as discussed in Ref. [1]). We consider only the case  $\mu > 1$ , since that is where the fixed point is unstable. When  $\Theta_n = 1$  (or 0), we say the control is active (or inactive) for one iteration. The problem is to understand which values of  $\beta$  will render the fixed point at  $X^*$  stable.

If the  $\Theta_n$  factor is removed from Eq. (2) so that control is always active, the method is identical to the discrete time-delay feedback discussed in Ref. [2]. In this case, it is known that the control is effective only for  $(\mu - 1)/2 < \beta < 1$ . (See Ref. [2], however, for a discussion of how this domain can be enlarged using a simple recursive algorithm.) Reference [1] notes that the introduction of the  $\Theta_n$  factor doubles the lower limit on  $\beta$ , but does not mention an important feature of the upper boundary.

Figure 1 indicates the full domain of stability of the fixed point of Eq. (1). It can be shown that only two types of sequence  $\Theta_n$  are possible as the system converges to the fixed point: periodic repetition of either 01... or 001..., where the dots indicate a string of  $k$  1's with  $k$  any integer. The infinite strips in Fig. 1 indicate regions where sequences with  $k = 1, 2$ , or 3 will be observed (in the absence of noise). The equations of the dark curves that bound the  $k = 1$  domain are  $\beta = \mu - 1$  and  $\beta = \mu - 1 + 1/\mu$ . In general, the boundaries of the domain of order  $k$  are solutions to polynomial equations of order  $k$ , so an analytic solution is possible only up to  $k = 3$  [3]. It can be shown, however, that the limit of the infinite sequence of stable domains is just the line  $\beta = \mu + 2 + 2\sqrt{\mu + 1}$ .

We note that the domain of control is actually enlarged by the inclusion of the  $\Theta_n$  factor, a surprising result given

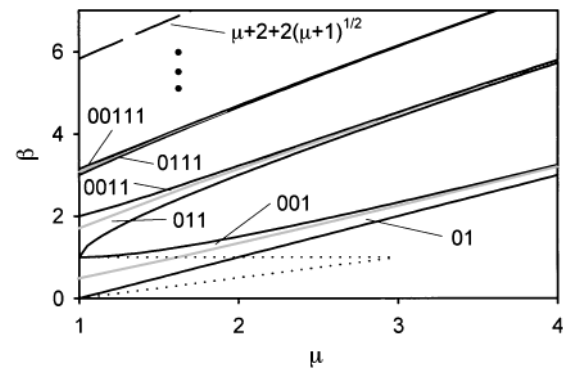


FIG. 1. The domain of control for Eqs. (1)–(3). The fixed point is stable within the regions labeled by the sequence of control activations. The light lines indicate optimal convergence. The large dots indicate that stable regions of higher order exist up to the limiting dashed line. The region outlined by dotted lines shows where the system would be stable without the  $\Theta_n$  factor [2].

that this factor models an *inability* to affect the system on certain iterations. In particular, the domain extends to arbitrarily large values of  $\mu$ , as compared to the case where control is always on for which  $\mu > 3$  cannot be controlled with any choice of  $\beta$ . We also note that optimal convergence rates are obtained by operating at any point on a boundary between the two sequence types of order  $k$ . At these points the system is directed to the fixed point after a single period of the control activation sequence. For  $k = 1$ , the boundary is given by  $\beta = \mu^2/(\mu + 1)$ .

Finally, we emphasize that control schemes in which feedback is based on comparison to previous states of the system rather than on an absolute reference seem to provide an efficient method of tracking the fixed point while system parameters slowly drift [1,2,4,5]. This makes a variety of methods based on time-delay feedback particularly interesting for biological systems, where appropriate response to parameter drift is often highly desirable.

This work was supported by NSF and the Whitaker Foundation.

Daniel J. Gauthier and Joshua E. S. Socolar  
Physics Department and CNCS  
Duke University  
Durham, North Carolina 27708

Received 25 August 1997 [S0031-9007(97)04701-7]  
PACS numbers: 87.22.-q, 05.45.+b, 07.05.Dz, 87.10.+e

- [1] K. Hall *et al.*, Phys. Rev. Lett. **78**, 4518 (1997).
- [2] J. Socolar, D. Sukow, and D. Gauthier, Phys. Rev. E **50**, 3245 (1994).
- [3] K. Hall (private communication) obtained these results independently.
- [4] D. Gauthier *et al.*, Phys. Rev. E **50**, 2343 (1994).
- [5] S. Bielawski, D. Derozier, and P. Glorieux, Phys. Rev. E **49**, R971 (1994).